

UNIVERSITY PARTNER



PROJECT PROPOSAL REPORT-

Digital Art Gallery Platform (A Platform to Showcase, Buy, and Sell Local Artworks)

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SECTION 1: STUDENT DETAILS

Academic Question: How can digital platforms enhance accessibility, engagement, and economic opportunities for local artists in developing regions like Nepal?

Aims:

- Build interactive online gallery for showcasing and selling local artworks
- Provide artist profile pages and portfolio management
- Enable customer-artist collaboration through commission requests
- Implement 3D virtual gallery browsing experiences
- Use AI recommendation systems for personalized suggestions
- Integrate secure backend for user and transaction management

Objectives:

- Design artist registration, authentication, and profile systems
- Develop artwork upload, management, and pricing features
- Create customer browsing, wishlist, and purchase modules
- Build admin dashboard for monitoring and analytics
- Integrate 3D visualization using WebGL/Three.js
- Implement AI-powered recommendations using Python
- Ensure secure data storage with MySQL
- Support responsive design for all devices

Artefact: Full-stack web platform using React.js (frontend), Node.js/Express (backend), MySQL (database), and Python (AI components)

SECTION 3: PROJECT PROPOSAL

3.1 Introduction

Artists in Nepal face significant barriers including high gallery costs (NPR 50,000-200,000 per exhibition), limited geographical reach, and lack of professional online platforms. Physical galleries are concentrated in urban centers, excluding rural

artists and limiting audience access. Artists currently rely on informal social media channels lacking professional presentation and secure transactions.

This Digital Art Gallery Platform addresses these gaps by democratizing art market access. Artists can create professional profiles, upload portfolios, and connect globally at minimal cost. The platform features 3D virtual galleries, AI-driven recommendations, commission systems, and secure payments, preserving cultural heritage through digital documentation.

I selected this project combining my interests in web development, AI, and cultural preservation. Nepal lacks specialized art platforms, creating opportunity for meaningful impact through technology that bridges traditional barriers while maintaining cultural authenticity.

3.2 Initial Research into Sources of Information

Munikumari, T., & Sai Kiran, P. (2024). Online Art Gallery. Details gallery management systems supporting diverse art forms, emphasizing database design for artist profiles and user engagement. Relevance: Guides database schema and interaction modules. Use: Inform artist profile systems and pricing mechanisms.

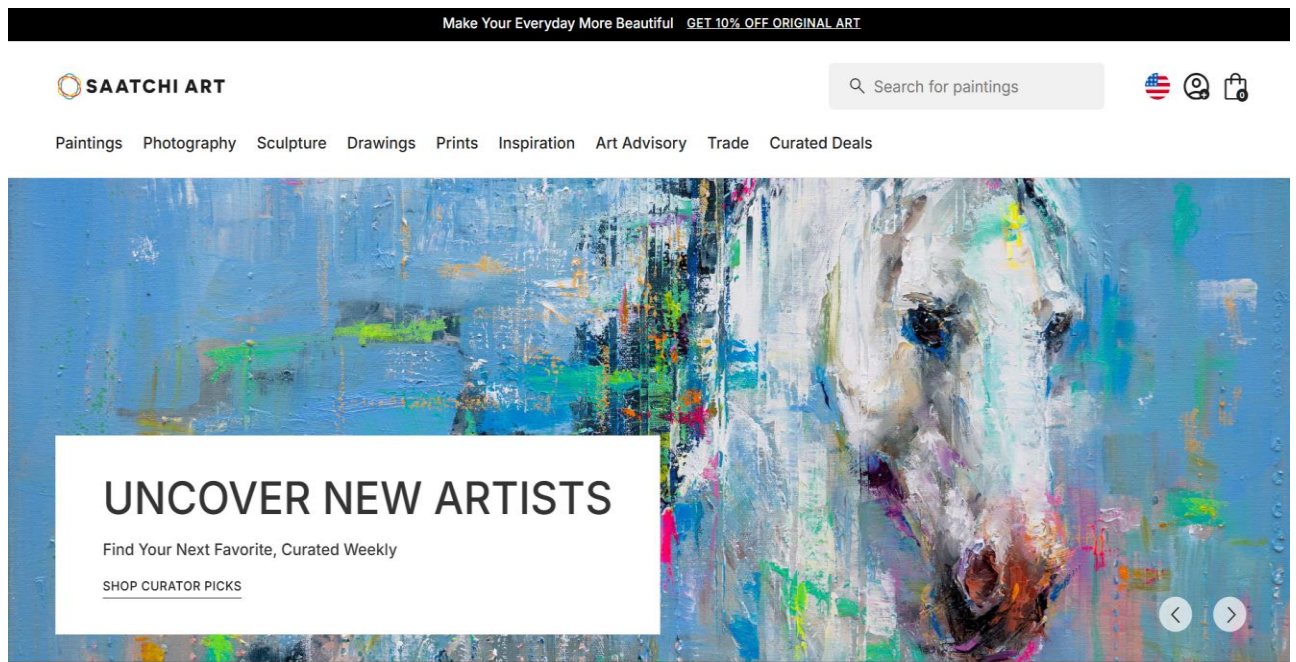
Rodriguez, R. M. (2024). The Psychology of Virtual Art Gallery Experiences. Explores emotional engagement in 3D galleries using OGAR framework and WebGL implementation challenges. Relevance: Supports 3D experience inclusion for enhanced engagement. Use: Inform 3D gallery design and user experience optimization.

Betrand, C. U., et al. (2022). Online Art Gallery Exhibition System for Indigenous Art Works. Proposes exhibitions/auctions for Nigerian artists using React.js/Node.js/MongoDB, emphasizing security and heritage protection. Relevance: Comparable to Nepal's context. Use: Inform security implementation and cultural preservation features.

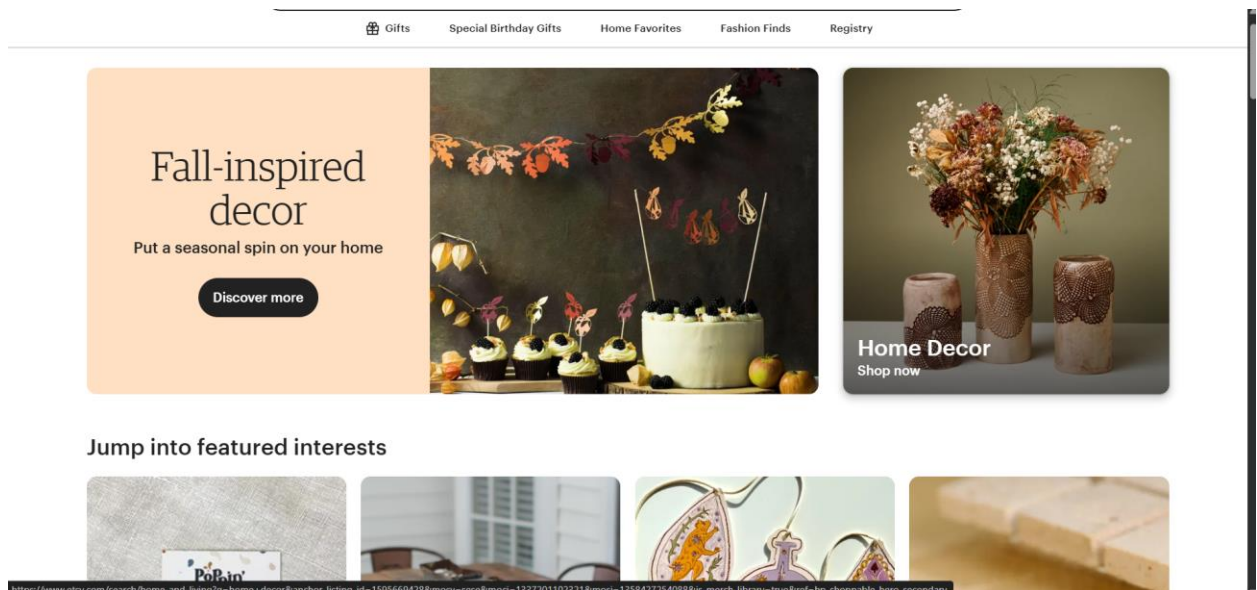
3.2.2 Market Analysis

Existing Online Art Platforms (International):

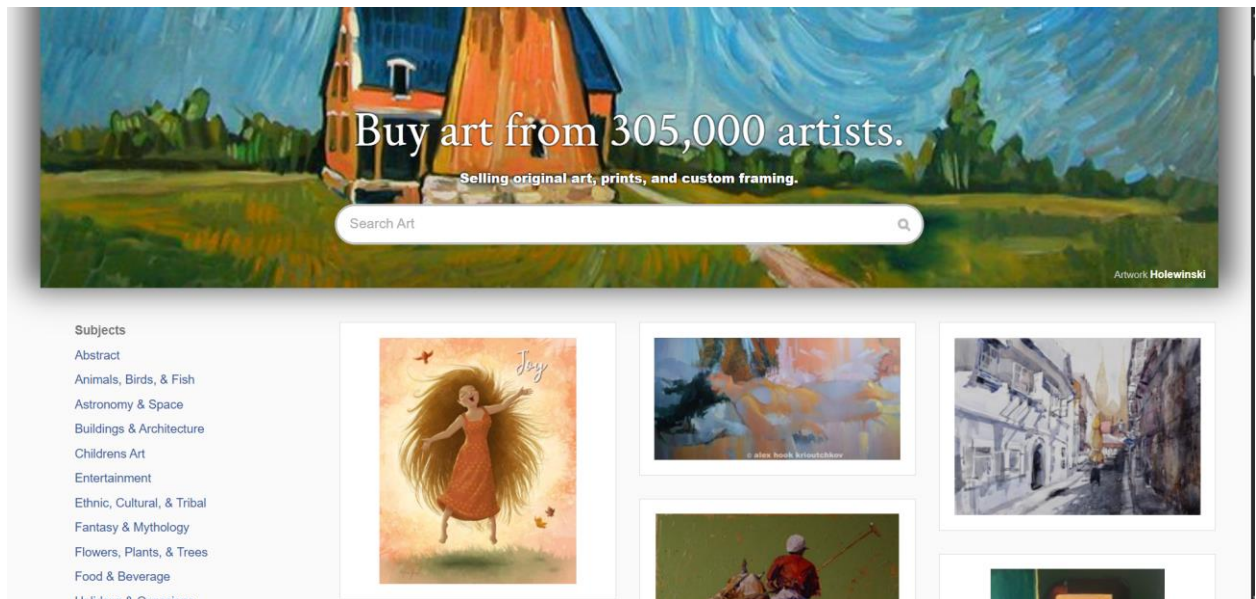
- **Saatchi Art:** World's largest online gallery enabling global artist sales, but charges high commission fees (35-50%), making it less accessible for emerging artists in developing regions.



- [Etsy](#): Supports handmade art sales but lacks dedicated 3D gallery experiences, AI recommendations, and specialized artist management tools.



- **ArtPal:** A growing online art platform that empowers artists with no commission fees, making it attractive for emerging creators. Artists can create a free online gallery, while ArtPal also offers print-on-demand services—handling printing and shipping for artworks—simplifying logistics for sellers. The platform is designed to be user-friendly, with social media integration allowing easy promotion. This makes ArtPal a strong model for art marketplaces that benefit artists directly.



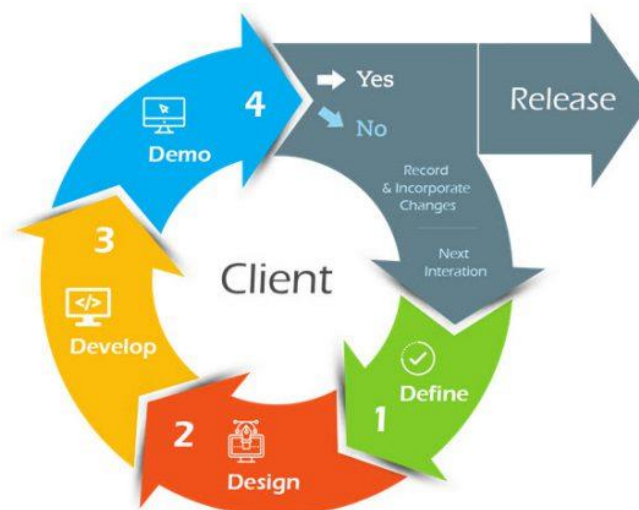
Local Scenario (Nepal):

- Nepal currently lacks a dedicated online art gallery platform specifically designed for local artists and cultural preservation.
- Artists rely on informal social media channels (Facebook, Instagram) which lack structured buying systems, professional presentation, AI recommendations, and 3D exhibition capabilities.
- This significant gap provides unique opportunity for the proposed platform to serve an underserved market while preserving cultural heritage.

3.3 Artefact (Proposed)

Justification: The web platform addresses the academic question by expanding artist opportunities through global exposure, immersive viewing, and structured management. Web-based ensures universal accessibility without installation barriers.

Methodology: Agile development for iterative feedback integration and flexible requirements. Development cycle: Requirements → Design → Backend → Frontend → AI Integration → 3D Integration → Testing → Deployment.



Technologies:

- React.js: Component-based responsive UI with excellent mobile support
- Node.js/Express: Scalable asynchronous backend with robust routing
- MySQL: ACID-compliant structured data storage with referential integrity
- Python/Scikit-learn: ML algorithms for recommendations with extensive documentation
- WebGL/Three.js: Cross-browser 3D graphics for virtual galleries
- AWS: EC2 hosting, RDS database, S3 storage, CloudFront CDN

Testing: Unit testing (Jest, 80% coverage), integration testing (API endpoints), user acceptance testing (local artists/customers), performance testing (3D rendering, mobile optimization), security testing (authentication, SQL injection prevention). Payment processing simulated using sandbox environments.

3.4 Plan/Schedule

The table outlines a 23-week project timeline for developing a Digital Art Gallery Platform, breaking the work into eight clear phases. Each phase has a start date, end date, duration, and key activities, helping everyone involved know what to focus on and when. The project kicks off on June 29, 2025, and wraps up by December 6, 2025, with specific tasks assigned to each stage to ensure steady progress.

The first few phases lay the groundwork. Phase 1 (Planning & Analysis) from June 29 to July 12 involves submitting a proposal, gathering requirements, and getting feedback from stakeholders. Phase 2 (System Design) from July 13 to July 26 focuses on creating a database schema, designing user interfaces (UI/UX), and planning the system architecture. Phases 3 (Backend Development) and 4 (Frontend Development) from July 27 to September 6 build the platform's core, with tasks like setting up APIs, authentication, database integration, and designing a responsive UI.

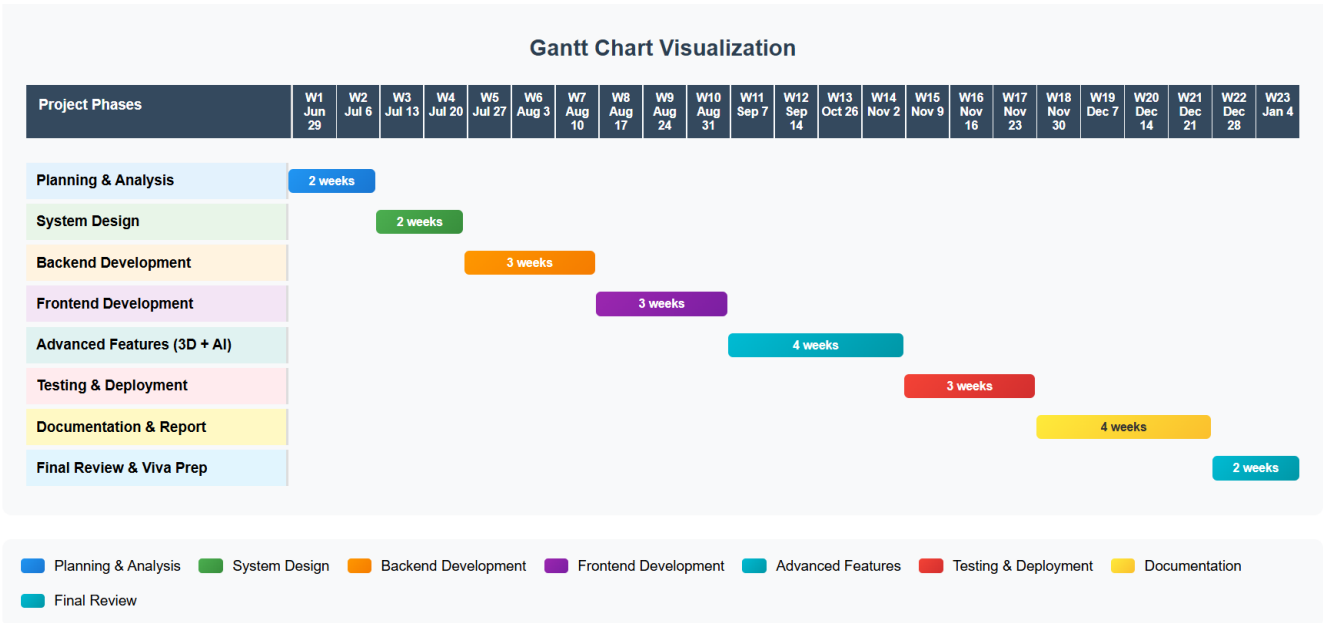
Later phases add advanced features and finalize the project. Phase 5 (Advanced Features) from September 7 to October 4 includes creating 3D virtual galleries, using tools like SQL and Three.js, and adding AI recommendations with Python. Phase 6 (Testing & Deployment) from October 5 to October 25 ensures everything works with comprehensive testing and AWS deployment. Finally, Phases 7 (Documentation & Report Writing) from October 26 to November 22 and 8 (Final Review & Viva Prep) from November 23 to December 6 involve writing reports, preparing a dissertation, and getting ready for a final review and presentation.

Digital Art Gallery Platform

23-Week Project Timeline with Exact Dates & FYP Milestones

Phase	Start Date	End Date	Duration	Key Activities
Phase 1: Planning & Analysis	Jun 29, 2025	Jul 12, 2025	2 weeks	Proposal submission, requirements gathering, stakeholder feedback
Phase 2: System Design	Jul 13, 2025	Jul 26, 2025	2 weeks	Database schema, UI/UX mockups, system architecture
Phase 3: Backend Development	Jul 27, 2025	Aug 16, 2025	3 weeks	RESTful APIs, authentication systems, database integration
Phase 4: Frontend Development	Aug 17, 2025	Sep 6, 2025	3 weeks	UI components, responsive design, frontend-backend integration
Phase 5: Advanced Features (3D + AI)	Sep 7, 2025	Oct 4, 2025	4 weeks	3D virtual gallery (WebGL/Three.js), AI recommendations (Python)
Phase 6: Testing & Deployment	Oct 5, 2025	Oct 25, 2025	3 weeks	Comprehensive testing, AWS deployment, performance optimization
Phase 7: Documentation & Report Writing	Oct 26, 2025	Nov 22, 2025	4 weeks	Literature review, dissertation writing, report preparation
Phase 8: Final Review & Viva Prep	Nov 23, 2025	Dec 6, 2025	2 weeks	Poster preparation, viva preparation, final review

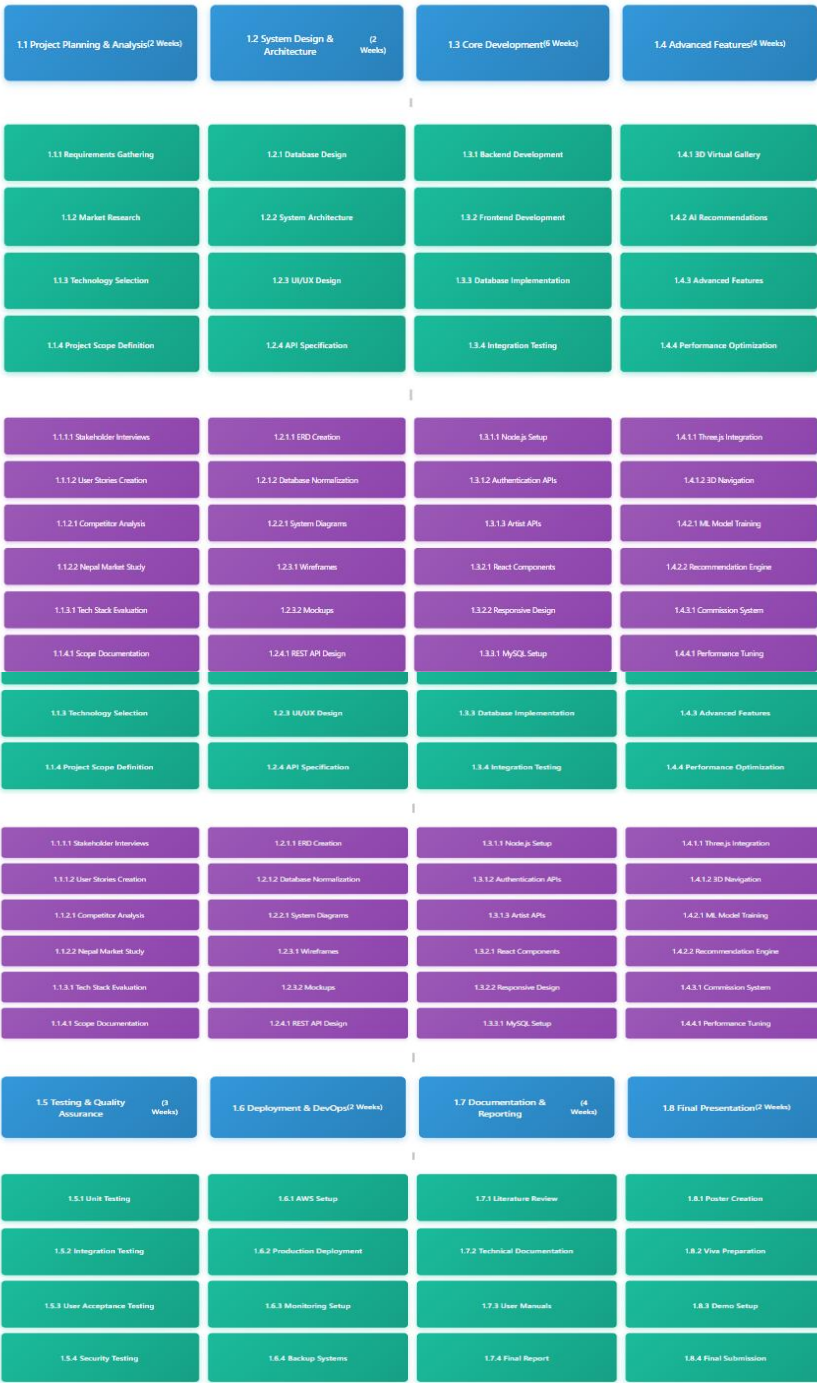
Gantt Chart Visualization



Work Breakdown Structure (WBS)

Digital Art Gallery Platform - A Platform to Showcase, Buy, and Sell Local Artworks

1.0 Digital Art Gallery Platform
(23 Weeks | React.js + Node.js + MySQL + AI)



3.5 References

[Betrand, C. U., Ekwealor, O. U., & Onyema, C. J. \(2022\). Online Art Gallery Exhibition and Auction System for Indigenous Art Works. International Journal on Data Science and Technology, 8\(3\), 55-60. Shodhshauryam, International Scientific Refereed Research Journal](#)

RODRIGUEZ, REBEKAH M. Ph.D. The Psychology of Virtual Art Gallery Experiences. (2024)
Directed by Dr. Paul J Silvia. 198 pp.

https://libres.uncg.edu/ir/uncg/f/Rodriguez_uncg_0154D_14030.pdf

Online Art Gallery Exhibition and Auction System for Indigenous Art Works

October 2022International Journal of Data Science Vol. 08(03):55-60([PDF](#)) [Online Art Gallery Exhibition and Auction System for Indigenous Art Works](#)

SECTION 4: ADDITIONAL INFORMATION

4.1 Resources

Hardware/Software: Standard laptop (8GB RAM), AWS free tier, MySQL Community Server

Technologies:

- Frontend: React.js (component architecture, virtual DOM), Tailwind CSS (responsive design)
- Backend: Node.js/Express (asynchronous handling), JWT authentication, bcrypt encryption
- Database: MySQL (ACID compliance, structured relationships)
- ML: Python/Scikit-learn (recommendation algorithms), Pandas/NumPy (data processing)
- 3D: WebGL/Three.js (cross-browser 3D graphics)
- Deployment: AWS EC2, RDS, S3, CloudFront
- Tools: Git/GitHub (version control), Jest (testing)

Algorithms: Collaborative filtering for recommendations, SQL transactions for secure payments, LOD algorithms for 3D optimization.

Conclusion

The proposed Digital Art Gallery Platform seeks to transform the way Nepalese artists present and sell their work by creating an accessible, professional, and secure online environment. At present, many local artists face high costs for physical exhibitions and limited reach due to the concentration of galleries in urban areas. This project addresses these barriers by offering a digital solution where artists can create profiles, manage portfolios, and showcase their artwork to a global audience. Key features such as AI-powered artwork recommendations, immersive 3D virtual gallery experiences, commission request options, and secure payment systems provide both innovation and reliability.

The platform will be developed using modern technologies including React.js for responsive design, Node.js/Express for backend services, MySQL for structured data management, Python for machine learning components, and WebGL/Three.js for 3D visualization. Hosting and deployment will utilize AWS cloud services to ensure scalability and performance. Built using an Agile methodology, the project emphasizes iterative feedback, testing, and adaptability throughout its 23-week timeline.

In conclusion, this platform not only creates new economic opportunities for local artists but also helps preserve Nepal's cultural identity in a digital form. It bridges the gap between tradition and technology, empowering creators to thrive in the global art market.